

Eugene Lafayette Buchanan

Senior AI Engineer — Agentic AI, MCP, RAG, LLM Routing

Apple Valley, CA (open to onsite Dearborn, MI) | San Quintin, Baja California, MX

+1 (909) 545 5384 | +52 (616) 126 5089 | eugene@serviceofothers.org | github.com/eugeneb50 — herdr-mcp MCP server

+ ZeroClaw agentic gateway (32.8k stars)

Professional Summary

Senior software engineer with 25+ years shipping production systems, including Product Owner, Test Automation Lead, and Senior Quality Assurance (QA) roles. Author of herdr-mcp, a Rust Model Context Protocol (MCP) server exposing the herdr agent multiplexer as 51 tools with an HTTP bridge and React playground. At Knowledgecity I helped ship a retrieval-augmented generation (RAG) AI support chatbot with guardrails under SOC 2 controls. I contribute agentic AI infrastructure to a 32.8k-star open-source Rust gateway (model routing, lifecycle observability, serving-provider usage identity). Enterprise integrations across SAP, Oracle, Workday, Coursera, UKG, and Zoom. Open to onsite in Dearborn, MI.

Core Skills — Mapped to TechDigital Requirements

MCP server, authored (core of this role): herdr-mcp in Rust — MCP stdio for Claude Desktop, Cursor, Claude Code, Continue, OpenCode

MCP tools and context management: 51 tools (discovery, lifecycle, read/write, sync, agent-to-agent, trim, recipes, scheduler, folder-key, clipboard); pane_id or label targeting; live agent registry over Unix socket events

MCP transports and UI: Axum HTTP bridge with JSON API; single-file React playground (run tools, chain recipes, inspect results); ratatui dashboard with mouse and keyboard nav

Agentic orchestration: recipe engine with variable interpolation; cron scheduler; agent lifecycle observer (idle/working/blocked/released); per-attempt usage ledger with serving-provider identity

RAG and search: shipped RAG chatbot with context retrieval and guardrails under SOC 2; Elasticsearch and SQL full-text search; evaluation of AI-generated code

Message compression: caveman (lossy) plus pfc1 (lossless phonetic) trim pipeline; per-pane policies; per-folder keys; trim analytics

LLM provider exposure: 9-provider routing via OpenRouter plus 8 OpenAI-compatible endpoints; per-dimension pricing with explicit unpriced-usage provenance

Languages and delivery: Rust, Python (Django/WordPress, evaluation scripting), JavaScript/TypeScript, Node.js, React, SQL; AWS (ECS pattern, S3); CI/CD gates; health dashboards; SOC 2; Blue Team forensics

Leadership: hired and trained automation team; Product Owner backlog with KPI tracking and PDR life cycles; cross-functional delivery with business stakeholders

Work Experience

Senior QA Engineer / Test Automation Lead / Product Owner — Integrations (incl. RAG chatbot)

Knowledgecity LLC, Remote, Dec 2020 – Aug 2025

- Support team needed grounded answers over scattered content: helped ship a RAG chatbot with context retrieval and guardrails that cleared SOC 2 review.
- Six platforms, six auth dialects: shipped SAP, Oracle, Workday, Coursera, UKG, Zoom integrations over SAML, OAuth, SFTP, REST APIs, webhooks, SCORM, LTI.
- Complex deployments needed fast feedback: ran CI/CD quality pipelines with Cypress, Selenium, Postman, JUnit, Elasticsearch, S3 plus health dashboards and alerting.
- Inherited thin automation coverage: hired and trained the test team and built end-to-end regression suites.
- Owned delivery as Integrations Product Owner: ran backlog and PDR life cycles with KPI tracking.

Author — herdr-mcp MCP Server (github.com/eugeneb50/herdr-mcp)

Independent, Rust, 2026 – Present

- Agents had no MCP path into herdr workspaces: authored a Rust MCP server exposing the multiplexer as 51 tools.
- Browsers needed access without a dev server: added an Axum HTTP bridge serving a single-file React playground for tools and recipes.
- Multi-step agent work needed repeatability: built a recipe engine with variable interpolation plus a cron scheduler.
- Agent-to-agent messages wasted context: added a caveman/pfc1 trim pipeline with per-pane policies and per-folder keys.
- Result: 4-crate workspace with 299 passing tests, working from Claude Desktop, Cursor, Claude Code, Continue, and OpenCode.

Open-Source Contributor — Agentic AI Infrastructure

ZeroClaw Labs, github.com/zeroclaw-labs/zeroclaw, 32.8k GitHub stars, 2026 – Present

- Routing and cost data scattered across 9 providers: built one context-window source of truth (PR #7946, merged Jul 2026, 39 commits).
- Fallbacks misattributed usage and cost: shipped live serving-provider identity on usage events with a per-attempt ledger and recovery policy (PR #8966, merged Sep 18, 2026, 94 commits).
- Agent state invisible to operators: built a lifecycle observer over JSON-RPC whose requirements fed the vendor-neutral design (PR #8337, superseded by #10269).

Senior Software Engineer II (SDET Practice)

RealNetworks, Seattle, WA, 1999 – 2001

- SDET quality assurance across Linux, Unix, Windows, and embedded systems for streaming media.
- Global presales engineering across Japan, Korea, and the US; trained new hires.

Earlier Software Test Engineering and Support Roles

Microsoft, IBM, Keene Inc., 1996 – 1999

- Microsoft (1999): Senior Test Engineer IV, Windows Media Server 4/5; nightly build automation.
- IBM (1998): Senior Test Engineer III, high-availability servers; WHQL cluster failover.
- Microsoft (1997) / Keene (1996): OEM setup verification; frontline and escalated support ticketing.

Business Consultant — Campus IT, Software Evaluation and Migration

Medico Investments LLC / Foremost Senior Care / Foremost Organization, Hesperia, CA, 2013 – 2016

- Evaluated and migrated accounting and medical-records software.
- Ran campus IT support: security/backup policies, surveillance, wireless, events.

IT Manager; Freelance IT Consultant (Django / WordPress); Retail Service Desk

Lucerne Valley Unified School District (2002); Amp Computer, Apple Valley (2003 – 2011); Geek Squad, Best Buy, Victorville (2007)

- District support across four campuses (WAN/LAN, NetWare, Windows, Linux) with ETL for state reporting.
- Freelance Django/WordPress builds with ERP, CRM, and office automation.
- Retail service desk: ticket triage, PC setup, data migration, hardware repair.

Education

- Associate Degree, Victor Valley College
- California Notary Commission
- Toastmasters International (technical communication and documentation)